

The Way to a Green Future

Though India's per capita CO₂ emission is one of the lowest in the world, it must carefully phase down thermal energy while reducing the carbon footprint

AT THE RECENT United Nations COP28 climate summit in Dubai, two terms took centre stage in discussions: mitigation and adaptation. The former refers to efforts made to decrease carbon emissions, urging nations to commit to reducing the release of greenhouse gases. The latter focusses on the adjustments that countries need to make to address these transformations.

Though it has one of the lowest per capita CO₂ emissions and energy consumption rates in the world, India has done remarkable work on both counts. Qatar has the highest CO₂ emissions per capita at 36 tonnes per person; for the US, the figure is around 15 tonnes per person, while in India it is only 2 tonnes, much lower even than the world average of 5 tonnes. In simple terms, it means that the average Indian household consumes only a tenth of the electricity consumed by an average household in the US.

Notwithstanding this, India has not only met but also exceeded its commitment made at the COP21 summit held in Paris in 2015 by generating 40% of its power from non-fossil fuels. The achievement is more impressive if one were to consider the strong relationship between income, health, water, infrastructure, education, life expectancy, and



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energy consumption per capita. India's proactive approach towards sustainable energy sources has allowed it to surpass expectations almost a decade ahead of schedule.

These climate initiatives were recognised as the fourth strongest globally in the latest Climate Change Performance Index brought out by the NGO Germanwatch e.V., which assesses the performance of countries on four indicators—total emissions, renewable energy deployment, energy use, and climate policy. In fact, India outperforms every major economy.

India is now the most populous nation with 1.4 billion people, almost a sixth of the world's population of 8 billion, and its energy needs are increasing at a fast clip, more than that of any other country in the coming decades. Peak power demand in India touched 239.9 GW on September 1, 2023, much more than the Central Electricity Authority's projection of 230 GW. This forces the country to rely on fossil fuels for power generation even though it has an installed non-fossil fuel capacity of 179.33 GW, with an additional 128 GW under development or bid out.

Adani Green Energy (AGEL) has a locked-in growth trajectory up to 20.4 GW and an operating renewable portfolio of 8.4 GW, the largest in India. AGEL's operating portfolio has cumulatively offset over 41 million tonnes of CO₂ emissions.